Multilingual Government Services Assistant

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Abstract

Accessing and understanding Indian government schemes remains a major challenge for citizens due to the use of complex official language, fragmented information sources, and widespread multilingualism. This study introduces YojnaConnect, an intelligent chatbot system that simplifies the process by enabling users to query government schemes in their preferred regional language, including code-switched inputs. Leveraging pre-trained NLP models from Hugging Face and Gemini 2.5 APIs for translation and summarization, the system performs semantic retrieval from a curated dataset of over 1,400 government schemes across states. The methodology includes data preprocessing, exploratory analysis, feature extraction (TF-IDF and bigrams), and semantic encoding using SentenceTransformers to match user intent with the most relevant schemes. A React Native frontend and Flask backend create a responsive, real-time interface where users can input natural language queries, select their state, and receive concise, personalized responses in their language. Experimental results demonstrate high retrieval accuracy and improved accessibility, making policy information more inclusive. Future enhancements will focus on integrating local language models and adaptive feedback mechanisms to further enrich user interaction and regional reach.

Introduction

With rapid technological advancements in India, access to government services is becoming smarter and more efficient [1]. However, many users still struggle to understand government schemes [2] due to the complex, technical language of official documents. In a linguistically diverse country like India, where people often use regional languages [3] and code-switch between languages, current systems are limited in handling such queries effectively. This research proposes developing an intelligent system that accepts user queries in their preferred language and retrieves relevant government scheme information from a selected state. The system leverages pre-trained NLP models from Hugging Face for document retrieval and addresses the challenges of multilingualism and code-switching using translation APIs. By focusing on accurate intent detection and intuitive responses, the system aims to make access to government scheme information [4] more inclusive and user-friendly.

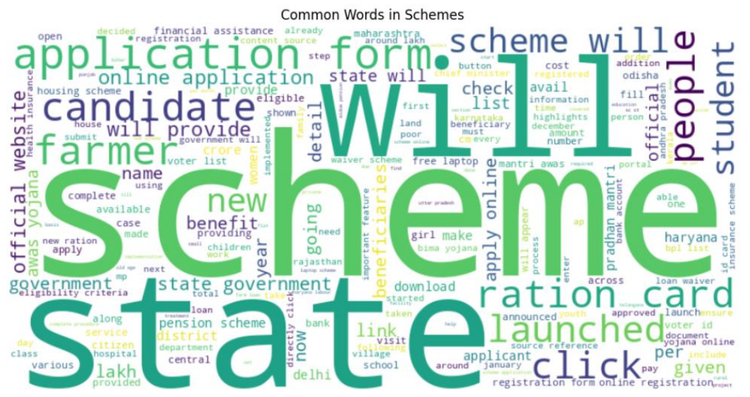

Figure 1. *A word cloud depicting the common words in the schemes*

Methodology

This research follows a structured methodology for developing a multilingual intelligent assistant to retrieve government scheme information. The dataset, sourced from Kaggle, includes 1,444 schemes across 29 directories, each in .txt format with key details like eligibility, ministry, and sector. It was transformed into a structured dataframe for analysis. Exploratory Data Analysis (EDA) included evaluating document distribution and length via bar graphs, histograms, and box plots. Preprocessing involved lowercasing, noise removal, tokenization, stopword removal, and stemming/lemmatization. Feature extraction techniques included TF–IDF, bigram frequencies, and semantic embeddings using the SentenceTransformer model (all-MiniLM-L6-v2). For semantic retrieval, user queries were embedded and matched to schemes using cosine similarity. Multilingual input was handled using Gemini 2.5 for translation and summarization, ensuring responses are concise and in the user's preferred language. The backend, built with Flask, processes input queries and returns summarized results. The frontend, developed with React Native and Expo SDK, allows users to input queries, select a state, and view results. Communication between the frontend and backend is enabled via Axios, with real-time testing and a feature to view past interactions. The system supports future deployment on platforms like AWS or Heroku.

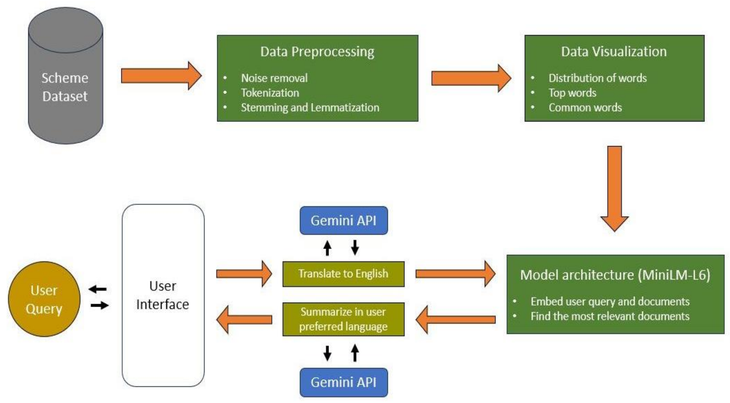

Figure 2. *Flowchart of the methodology*

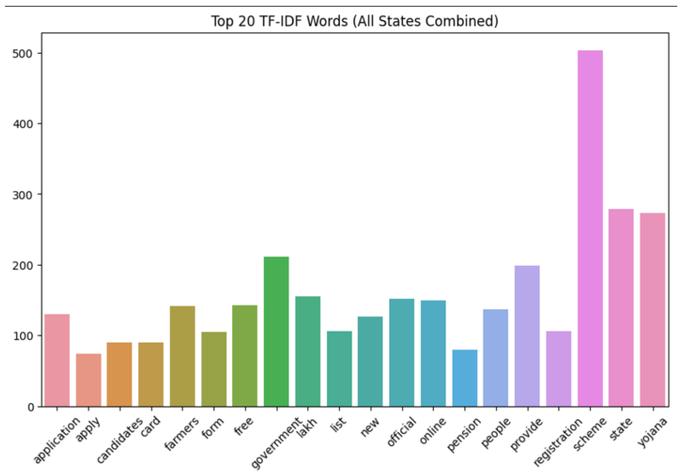

Figure 3. *A bar graph depicting the top 20 TF-IDF words*

Conclusion

This study presents a comprehensive pipeline for organizing, analyzing, and retrieving Indian government scheme documents using modern NLP techniques. By combining statistical text analysis, semantic search through transformer-based embeddings, and abstractive summarization via a generative API, the system effectively addresses the challenges of document heterogeneity, multilingual content, and information overload. The strong retrieval performance and high-quality summaries demonstrate the potential of such approaches to improve accessibility and understanding of public policy content. Future work may explore integration with regional language models and user feedback mechanisms to further enhance relevance and inclusivity.

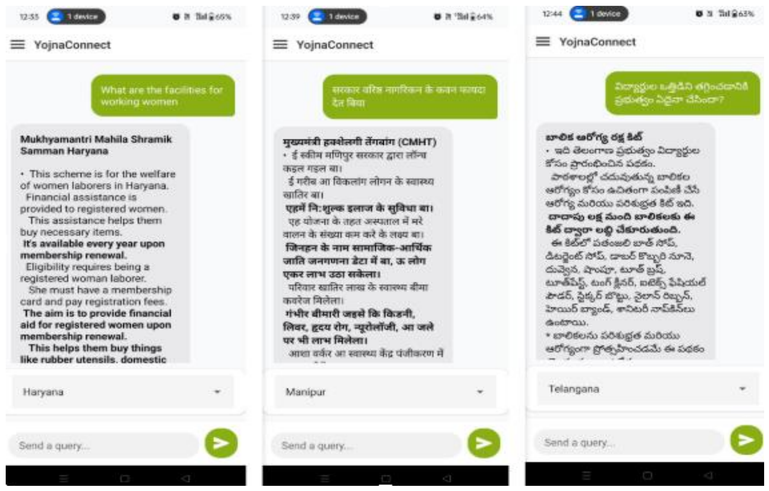

Figure 4*. App Preview*

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